Supplementary material. Full technical report

Incremental predictive value and perturbation-theoretic validity of a spectral fragility index for atrial reentry

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This supplement is the complete technical report behind the main article. It contains the full methods, the substrate provenance audit and its four controls, the independent-solver cross-check, the resolution-convergence study, the cross-medium localization results, the power analyses, the formal (proof-assistant) verification of the algebraic core, and the dated record of every correction made during this work. Every number in the main article appears here with its derivation and its stored source file.
Sections are numbered independently of the main article. References to `results/*.json` are files in the public project repository.

Abstract

We ask whether a closed-form spectral graph biomarker predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. The biomarker is the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity $\partial\lambda_2/\partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$ aggregated under a fibrosis-weighted conduction-uncoupling field. Pre-registered before any labels were seen, the primary endpoint was $\Delta\text{AUC} \geq 0.05$ with a DeLong paired $p < 0.05$ on shape-family-held-out folds. The endpoint is not met, but the strength of the negative differs by regime, and we report the two regimes separately. On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace $\Delta\text{AUC} \approx +0.003$; $\sim$4,150 cases needed at 80% power, resolvable only because N is huge, and so clinically undetectable there). On two real cohorts (182 patients: Roney $n = 100$, UW/Boyle $n = 82$) the combined gradient-boosting estimate is $+0.012$ with a bootstrap interval $[-0.017, +0.044]$ that excludes the threshold, so the real-anatomy result is a measured null and not a shortfall of power. An earlier version of this work reported $+0.051$ here and called the real-cohort evidence inconclusive. Correcting how we had read one cohort's released substrate, which we found only afterwards, moved that figure to $+0.036$ at the original cohort size, and extending the Roney arm from 62 to all 100 released meshes moved it to $+0.012$, so most of the drop was sample size, not substrate. The audit that found the misreading, and the four controls that followed, are reported in full; the cause of that cohort's unusually low inducibility rate proved to be a 25% difference in fibrosis burden between the two released cohorts rather than any substrate defect, and that difference accounts for the apparent anomaly and reproduces it in either direction when the burden is swapped. Where the null is clean, only half the mechanism is feature redundancy: the aggregate SFI summaries are reconstructible from the competitor set at $R^2 = 0.970$, while the extreme and edge-level columns retain variance of their own ($R^2 < 0.25$) and still do not predict the label. The exact per-region $\Delta\lambda_2$ SFI, which has no first-order approximation error, also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is also ill-conditioned on real tissue: it is valid only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$, and real atria have median $\rho \approx 2422$, at least $272\times$ past the empirical $\rho^* \approx 3$ on every mesh tested, so by Davis–Kahan the Fiedler vector is unstable. We do *not* claim ρ proves non-predictiveness; it bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a ranking feature. The spectral gap of the weighted conduction Laplacian collapses as $N^{-1.00}$ on the real atrial surface, matching the 2-manifold Weyl exponent, so ρ grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer $|\nabla\varphi_2|$ is a weak but real rank localizer (origin in the top $\sim 18\%$ on synthetic networks, and weaker on real anatomy, where only ~ 20 origins are available) that beats graph-centrality baselines and transfers across three dynamical media, though it co-localizes with substrate imaging rather than beating it, and substrate imaging is the better localizer in the FHN arm ($\sim$0.10 against 0.176), for a reason we make precise. Naive methodology (random CV, p-value thresholds, no effect-size gate)

would have reported SFI as working, whereas our pre-registered protocol rejects it. Our own convergence study identifies a key limitation: the inducibility labeller is strongly resolution-sensitive and, on a completed six-tier study (1500–48000 nodes, $n = 24$), does not converge within the tested window; the direction of the residual bias at the 2,000-node operating resolution is not established, since the finest tier (37.5%) sits above the operating-resolution rate (34.0%). A pre-registered clinical endpoint on two-year recurrence outcomes for the 82 UW patients supplies the study's sharpest illustration: its gate fired, at $\Delta\text{AUC} = +0.197$ with $p = 0.0061$, and the analysis plan frozen before the outcomes were joined identifies the result as an artefact of an anti-predictive baseline (AUC 0.250) — an increment that nine columns of random noise reproduce (empirical $p = 0.333$); the endpoint is not met. Our contribution is a carefully bounded negative result: the mechanism behind it, a label-free criterion for the estimator's validity, a characterisation of how it scales, and a demonstration of the false positives that standard methodology produces, including one this protocol caught on real patients.

Keywords: algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

Ground-truth labels are electrophysiology-simulator inducibility verdicts throughout, with one exception: the pre-registered clinical endpoint of Section 4.2, which is scored against real two-year post-ablation recurrence outcomes for the 82 *UW patients. Every claim outside that section is in-silico. All numbers trace to* `results/*.json` *and* `notebooks/lab_notebook.md` *in the project repository.*

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1 Introduction

Algebraic connectivity λ_2 (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks (the master-stability-function result [3] and the synchronizability eigenratio λ_N/λ_2 [4]), and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \tag{1}$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis, and we pre-register the null ("SFI merely re-encodes fibrosis") as an accepted, reportable outcome, so that a negative result counts as a success of the protocol and not as a file-drawer casualty.

The mathematics we use is classical, and we claim no new theorem. Our contributions are (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a label-free validity criterion and a measured gap-scaling exponent that matches the Weyl 2-manifold prediction on real atria, which together explain why the *single-vector estimator* is ill-conditioned on real tissue and why any first-order Fiedler-based marker faces the same conditioning problem there, and which do not explain the predictive null; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

In brief, a closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features, either at synthetic scale or on 182 patients of real anatomy. The mechanism is partly feature redundancy and partly a feature that is genuinely new but unpredictive (Section 4.3). Independently of that null, the estimator's own conditioning fails there ($\rho \gg 1$). What we report is a carefully drawn boundary rather than a discovery: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

2 Theory: the Spectral Fragility Index

The core mathematics is classical numerical linear algebra, and the contribution lies in its use: each result is stated together with its prior art (Section 2.5), and the validity boundary is made explicit (Section 2.4).

2.1 The weighted conduction graph

Nodes are mesh vertices and edges are triangle edges, with the edge weight $w_{ij} > 0$ a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \qquad D = \operatorname{diag}\Big(\textstyle\sum_j w_{ij}\Big). \tag{2}$$

L is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum $0 = \lambda_1 < \lambda_2 \le \lambda_3 \le \cdots$ and eigenvectors $\varphi_1, \varphi_2, \ldots$; λ_2 is the algebraic connectivity and φ_2 the Fiedler vector.

2.2 The closed-form fragility (first order)

L is a sum of rank-one edge terms, so

$$L = \sum_{(i,j)\in E} w_{ij}\,(e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \tag{3}$$

For a *simple* eigenvalue λ with unit eigenvector φ, first-order (Hellmann–Feynman) perturbation theory gives $\partial\lambda/\partial w_{ij} = \varphi^\top(\partial L/\partial w_{ij})\varphi = (\varphi_i - \varphi_j)^2$, which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance w_{ij} by a small $\Delta w_{ij} \ge 0$ (uncoupling) lowers λ_2 by $\Delta w_{ij}(\varphi_{2,i} - \varphi_{2,j})^2$ to first order.

The identity is verified in six independent ways. In computer algebra, (i) sympy proves Eqs. (1)–(3) symbolically, and (ii) on a concrete 5-node graph the finite-difference $\partial\lambda_2/\partial w_{ij}$ matches $(\varphi_{2,i} - \varphi_{2,j})^2$ to $|\mathrm{err}| \approx 2 \times 10^{-17}$ (mpmath, 40 digits). Exactness and implementation independence are checked separately: (iii) the identity holds exactly with rational weights (zero floating-point error); (iv) an interval-arithmetic enclosure certifies the true λ_2 to within $\sim 4 \times 10^{-40}$; and (v) λ_2/φ_2 agree across scipy, networkx, and a hand-rolled inverse iteration (spread 2.5×10^{-15}). These three tolerances are printed at run time by `scripts/verify_sfi_identity.py` and `scripts/verify_math_rigor.py` rather than stored in `results/`, so they are reproduced by rerunning those scripts. Finally, (vi) the algebraic core is kernel-checked in the Coq proof assistant over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions Δw (mean $\mathbb{E}[\Delta w_{ij}] = \mathrm{frac}_{ij}\, w_{ij}$, calibrated to literature perioperative conduction-velocity slowing; $\Delta w \approx 0.36$, frozen in the pre-registration). The per-region Spectral Fragility Index is the expected λ_2 drop contributed by region R:

$$\mathrm{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j)\in R} \mathbb{E}[\Delta w_{ij}]\,(\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} - \underbrace{\sum_{k\ne 2} \frac{(\varphi_k^\top \Delta L\, \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{second-order resolvent}} + O(\|\Delta L\|^3). \tag{4}$$

The resolvent term is *subtracted* because the left-hand side is a drop, under $L' = L - \Delta L$; since every denominator $\lambda_2 - \lambda_k$ with $k \ge 3$ is negative and the $k = 1$ term vanishes (ΔL annihilates the constant vector φ_1), that subtracted sum is itself negative, so the true drop *exceeds* the first-order estimate — which is what the validity sweep measures, with the exact $\Delta\lambda_2$ above the first-order prediction at every ρ. The second-order term is the source of the validity radius developed below. Near eigenvalue degeneracy ($\lambda_2 \approx \lambda_3$) the single φ_2 is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with

$P = \sum_c \varphi_c \varphi_c^\top$ over the clustered subspace, $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$ ($s_{ij} = e_i - e_j$) is well-defined. For discrete edits (Maze lesion, atriotomy) the expansion is abandoned and λ_2 is recomputed exactly after the rank-one edit.

2.4 The validity radius: the central caveat

The first-order form is trustworthy only while the perturbation is small relative to the spectral gap $\lambda_3 - \lambda_2$. Weyl's inequality and the Davis–Kahan theorem bound different quantities, and applying Weyl to both is a common error:

- Weyl's inequality bounds the *eigenvalue* move: $|\Delta\lambda_2| \le \|\Delta L\|$.
- Davis–Kahan [7] bounds the *eigenvector* move, which scales like $\|\Delta L\|/(\lambda_3 - \lambda_2)$; this ratio, and not the Weyl bound, governs when φ_2 (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \tag{5}$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order $\Delta\lambda_2$ prediction is $\Theta(\rho)$, and the Fiedler vector rotates by $\sin\angle(\varphi_2, \varphi_2') \le \sqrt{2}\,\rho$ (Davis–Kahan). Empirically the error is 1% at $\rho = 0.3$, crosses 10% at $\rho^* \approx 3$, and reaches 92% at $\rho = 28$, with Weyl holding throughout. On real excitable media the spectral gap is small and ρ is correspondingly large: over the 12 scanned meshes the median is $\rho \approx 2422$ with an interquartile range of [1082, 7384] and a full range of [817, 16554]. The spread is wide because ρ inherits the near-degenerate gap in its denominator, but every mesh exceeds ρ^* and even the smallest does so by a factor of 272, so the conclusion does not rest on the median, which itself sits about $10^3\times$ past ρ^*.

The criterion ρ proves that the *single-vector* first-order SFI is a numerically invalid estimator of $\Delta\lambda_2$ on real tissue, and that φ_2 is ill-conditioned (Davis–Kahan), a label-free statement about the estimator. It does *not* prove SFI is non-predictive: ρ bounds the accuracy of the $\Delta\lambda_2$ magnitude, not the classification content of a monotone ranking feature; indeed the exact $\Delta\lambda_2$ SFI (no ρ limitation) is also non-predictive (Section 4.1), so the predictive null has a separate cause, which Section 4.3 measures and finds to be only half redundancy. The label-free status of ρ is only partial, since its numerator carries the fibrosis-weighted Δw field and only its denominator $\lambda_3 - \lambda_2$ is truly label-free; and because $\rho^* \approx 3$ is calibrated on a 16-node path graph with a problem-dependent constant, "$10^3\times$ past ρ^*" is order-of-magnitude, not exact.

2.5 Prior art: the mathematics is not new

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1]; λ_2 governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl's law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that ρ is the right *a priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to mechanistically explain a negative result.

3 Methods

Full detail is in the pre-registration (`docs/PRE_REGISTRATION.md`) and theory note (`docs/SFI_THEORY.md`); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

Reporting guideline. The pre-registered clinical analysis of Section 4.2 — the prediction of two-year post-ablation arrhythmia recurrence in the 82 University of Washington patients, and

the only analysis in this work scored against a real patient outcome — is reported in accordance with the **TRIPOD+AI** statement (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis, updated for prediction models developed with regression or machine learning methods) [19]. A completed TRIPOD+AI checklist accompanies this manuscript as a supplement. TRIPOD+AI supersedes the original TRIPOD statement, which its authors state should no longer be used, and it is the applicable version here because the clinical analysis reports both a penalised logistic regression and a gradient-boosted tree ensemble.

The scope of that adherence is stated deliberately rather than left to inference. Every other endpoint in this paper is in silico: the outcome is an electrophysiology simulator's reentry-inducibility verdict, not a patient outcome, so those analyses fall outside TRIPOD+AI's remit and no claim of adherence is made for them. They are nonetheless reported against the same frozen protocol, and the checklist marks each item accordingly instead of recording compliance the design cannot support. STARD was considered and judged inapplicable: this work evaluates the incremental discrimination of a multivariable prediction model, not the accuracy of an index test against a reference standard, and reports no sensitivity, specificity, or predictive values at a diagnostic threshold.

The edge weight. Everything downstream (λ_2, the Fiedler vector, the spectral gap, ρ and the SFI itself) is a function of the conduction graph's edge weights, so we give their form explicitly. For an edge (i, j) let $\hat{e}_{ij}$ be the unit vector along it, $\bar{f}_{ij}$ the normalised mean of the two endpoint fibre directions, and $a_{ij} = |\hat{e}_{ij} \cdot \bar{f}_{ij}|$ their absolute alignment. With $\phi_{ij} = \frac{1}{2}(\phi_i + \phi_j)$ the mean endpoint fibrosis,

$$w_{ij} = \underbrace{\left[c_\perp + (c_\parallel - c_\perp)\, a_{ij}^2\right]}_{\text{anisotropy}} \cdot \underbrace{\operatorname{clip}(1 - \phi_{ij},\, \varepsilon,\, 1)}_{\text{fibrosis}}, \tag{6}$$

with $c_\parallel = 1.0$ along fibre, $c_\perp = 0.3$ across, and a floor $\varepsilon = 0.05$ so dense scar retains a small residual conductance instead of disconnecting the graph. The a^2 dependence makes the weight a function of alignment and not of direction, so it is invariant to fibre sign and depends only on alignment, which is the only property of $\bar{f}_{ij}$ that is identifiable. Both factors are bounded, so $w_{ij} \in [\varepsilon c_\perp,\, c_\parallel]$; no edge weight can vanish and the graph stays connected by construction.

Cohort flow. The Roney deposit contains 100 meshes and all 100 are used. The UW deposit contains 164 meshes, which are 82 patients imaged twice, before and after ablation; the primary analysis uses the 82 pre-ablation meshes only, since post-ablation geometry encodes the delivered treatment. No subject is excluded for any other reason, so the analysed cohort is $100 + 82 = 182$ and there is no attrition to report. The 164-mesh figure appears in this paper only where a statement concerns the released files themselves, such as the fibre-field and element-tag audits.

Inducibility protocol. Labels come from a monodomain Mitchell–Schaeffer solve on the coarsened surface (2000 nodes, vertex clustering). Membrane parameters, frozen before any label was generated: $\tau_{\text{in}} = 0.3$, $\tau_{\text{out}} = 6.0$, $\tau_{\text{open}} = 120$, $\tau_{\text{close}} = 110$ ms and $v_{\text{gate}} = 0.13$, with fibrosis shortening τ_{close} by up to 50%. Diffusion uses an anisotropic conductance-weighted cotangent Laplacian with $d_0 = 0.20$ and $w_{\text{ref}} = 0.3$, integrated by backward Euler at $\Delta t = 0.05$ ms. Induction is burst pacing from 2 randomly chosen sites, 6 beats at a cycle length of 150 ms, stimulus amplitude 0.15 over 2 ms within a 4 mm radius. A subject is labelled inducible if self-sustained supra-threshold activity persists at least 650 ms after the last stimulus, observed over a 1000 ms window. The protocol was verified before use: APD_{90} 218 ms healthy and 121 ms fibrotic, planar conduction velocity 0.4–1.2 m/s, and a monotone relationship between fibrosis and sustained reentry.

Evaluation pipeline. Cross-validation is `GroupKFold` with the group set to the patient, so no subject appears in both training and test folds, using $k = \min(5, n_{\text{groups}})$ folds. Of the two classifiers, logistic regression is a standardisation-plus-`LogisticRegression` pipeline (`liblinear`, at most 2000 iterations) whose regularisation C is chosen in an inner `GroupKFold` over $\{0.03, 0.1, 0.3, 1.0, 3.0\}$, so that tuning never sees the outer test fold, whereas gradient boosting is deliberately *not* tuned (120 trees, depth 2, learning rate 0.05, subsample 0.8) because at these sample sizes a search would fit the folds; it is a fixed, shallow reference and not a competitive model. We log it as a deviation from the pre-registered nested selection, which specifies an inner `GroupKFold` search for both classifiers. Out-of-fold predicted probabilities are pooled across folds and compared by the DeLong test. Pooling out-of-fold scores means the two compared vectors come from k different fitted models, not from the two fixed scoring functions that DeLong's test assumes; the paired structure and the leakage-free grouping are preserved, but the resulting p-values should be read as approximate. The naive comparison reported alongside replaces the grouping with shuffled `KFold`. On the real cohorts each patient contributes one row and is its own group, so the grouping constraint is never binding — no partition of the rows can place a patient in both training and test — and `GroupKFold` degenerates into a deterministic split; the grouping is a guarantee rather than a correction. (It does not reduce to *unshuffled* `KFold`: with equally sized groups scikit-learn assigns them round-robin, so the folds are strided rather than contiguous. The distinction does not affect any reported number, since no unshuffled-`KFold` arm is reported.) The leakage the grouping exists to prevent is demonstrated on the replicated synthetic cohort, where resamples of a single shape family have several members (Section 4.11).

Substrate. Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad `10.5061/dryad.kkwh70sg0`, 82 distinct AF patients [12]). The two encode fibrosis differently, which matters wherever their burdens are compared. Each Roney mesh ships an image-intensity-ratio (IIR) field, mapped to a continuous fibrosis fraction by the Khurram healthy-to-scar ramp [13]. The UW release ships no IIR (the intensity ratios are withheld under that cohort's IRB), so its fibrosis is derived instead from the per-element tissue tags, which the cohort's authors define as atrial non-fibrotic, atrial fibrotic, veins and valves, and ablation scar. On the pre-ablation meshes analysed here only the first three occur, so UW fibrosis is two-valued once the non-myocardial class is removed, whereas Roney's is continuous. The cohorts differ in what else they carry: the Roney meshes ship registered universal atrial coordinates and per-vertex fibre fields (directional spread 0.922–0.987), whereas the UW meshes ship no atrial coordinates, for which we substitute a PCA chart, and no fibre orientations, likewise withheld, so that the `VECTORS fiber` array they do ship is the constant $(1, 0, 0)$ everywhere (spread 0.000) and UW anisotropy is a fixed coordinate bias rather than anatomy (Sections 4.4 and 4.5). Meshes are coarsened to $\sim$2000-node graphs by field-preserving vertex clustering.

Labels. Ground truth is a monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry $\geq$ 650 ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength (CV$\times$ERP), source–sink mismatch, and unidirectional block, *never* by λ_2, so a predictive SFI$\rightarrow$label link is not circular. These verdicts are those of a monodomain simulator and not clinical post-operative AF. openCARP is run alongside on all 182 subjects as a robustness check rather than as the labeller (`results/opencarp_full_cohort.json`), and agrees with the monodomain verdict on 156/182.

The SFI feature and its competitors. Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against

a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2-alone, which isolates what the SFI construction adds over the bare algebraic connectivity.

Evaluation protocol. The pre-registered guards are shape-family `GroupKFold` (each patient is its own group, so cross-validation is patient-held-out, with grouped and naive AUC both reported so that leakage is visible), the DeLong paired test [8] in its fast midrank form [9], $\geq 10^4$ stratified bootstrap resamples (a logged deviation: the Kuramoto transfer arm was run at 5×10^3), a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Gating applies to the effect size as well as to the p-value: the primary endpoint requires *both* grouped $\Delta\mathrm{AUC} \geq 0.05$ and DeLong $p < 0.05$.

Synthetic media, generative model. The synthetic cohorts are random geometric graphs, which give a two-dimensional excitable sheet with the degree heterogeneity of real tissue and none of its anatomy. Each network draws n points uniformly on the unit square and connects every pair within a fixed radius; if that radius leaves the graph with no edges at all, the construction falls back to a symmetric 5-nearest-neighbour graph (each node joined to its five nearest neighbours, mean degree ≈ 6 after symmetrisation, though at the radii used this branch is never taken), and in either case only the largest connected component is retained, so the medium is a single sheet, not several. Lesion burden is swept uniformly over [0.05, 0.45] across the cohort so that both classes appear, the network being stable at the low end and unstable at the high end, mirroring the mild-to-severe fibrosis gradient of the atrial cohorts. Because the topology is generated rather than measured, these cohorts carry no fibre field and no anatomical coordinates, and they are used only where those are irrelevant: for the scale behaviour of the null and for the cross-medium transfer of the localizer.

Transfer media. The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where λ_2 governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

4 Results

4.1 The predictive null across cohorts and scale (primary endpoint not met)

Table 1: Incremental predictive value of SFI over the full competitor set (`results/gm1_expanded_metrics.json`; the cohort-size-matched comparison is `results/gm1_matched_n.json`). ΔAUC is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped $\Delta\text{AUC} \geq 0.05$ *and* DeLong $p < 0.05$) is met in no regime, and every interval on real anatomy contains zero. The UW rows rest on eight positives and are uninformative rather than null; their base AUCs differ sharply between the competitor and fibrosis-only baselines, and that difference inflates the latter's apparent increment.

cohort / medium	events/N	model	base AUC	SFI ΔAUC (95% CI); p
Roney (real)	34/100	lr	0.870	-0.021 $[-0.075, +0.033]$; 0.44
		gbt	0.856	$+0.009$ $[-0.029, +0.046]$; 0.62
UW/Boyle (real)	8/82	lr	0.834	$+0.000$ $[-0.073, +0.081]$; 1.00
		gbt	0.748	$+0.035$ $[-0.095, +0.203]$; 0.67
Combined real	42/182	lr	0.884	-0.007 $[-0.042, +0.029]$; 0.70
		gbt	0.851	$+0.012$ $[-0.017, +0.044]$; 0.45
FHN networks	2,000 networks			$\leq +0.003$; $p \geq 0.12$
FHN networks	up to 100,000			subspace $+0.0028$; $p < 10^{-4}$ at $N \geq 10^4$
Kuramoto networks	500 networks			$\leq +0.0036$; $p \geq 0.38$

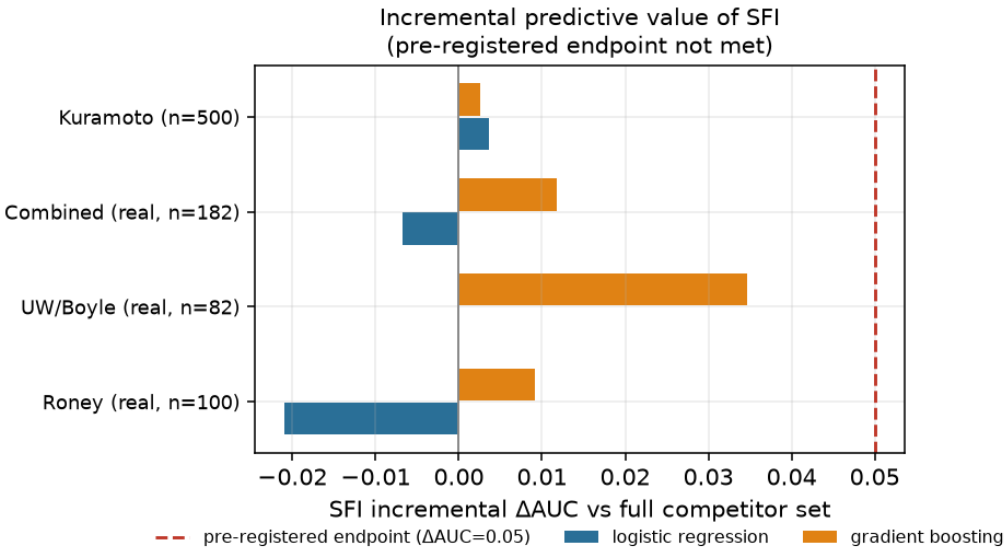


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort. On Roney and on the combined cohort the bootstrap interval excludes the gate as well as containing zero, so those are measured nulls and not a shortfall of power; the two UW intervals reach past the gate ($[-0.073, +0.081]$ and $[-0.095, +0.203]$ on eight positives) and are uninformative, not null.

The negative is not uniform across regimes. On the synthetic cohorts (up to 100,000 networks) the null is clean, the effect converging to $\Delta\text{AUC} \approx +0.003$ and becoming statistically resolvable only because N is enormous, whereas on the real cohorts it is a measured null: the combined gradient-boosting point estimate is $+0.012$ (DeLong $p = 0.449$, bootstrap CI $[-0.017, +0.044]$), roughly a quarter of the pre-registered 0.05 threshold and with an interval that excludes it. An

earlier version of this analysis reported +0.051 here and had to argue that the endpoint was missed only on significance. That reading does not survive, and the two changes behind its collapse weigh unequally. Correcting the sealed UW substrate while holding the cohort at its original size takes the estimate to +0.036 — already below the pre-registered gate, but with most of the old figure intact, so it was not principally a substrate artefact; extending the Roney arm to all 100 released meshes then takes it to +0.012. The larger share of the drop is sample size rather than substrate. The provenance audit that found the substrate misreading is Section 4.5, and the correction is set out alongside the other two in Section 5.1. With 42 inducible of 182 this is the largest real-anatomy arm in the study, and the null here is a measurement and not a shortfall. The combined row is a single pooled ROC across two cohorts of unequal prevalence (34/100 on Roney, 8/82 on UW), so its ranking carries between-cohort as well as within-cohort separability; the pooled base AUC under logistic regression, 0.884, consequently exceeds both components (0.870 and 0.834), and the pooled increment should be read alongside the cohort-specific estimates, which it does not replace.

The UW cohort remains the weak arm, though not for the reason we previously gave. On eight positives its intervals are wide, the competitor comparison spanning $[-0.095, +0.203]$ under gradient boosting, so it is an underpowered external check of the labeller rather than of SFI. We had also described its competitor baseline as close to chance, and that was simply wrong: the measured competitor base AUC is 0.834 under logistic regression and 0.748 under gradient boosting. Only the *fibrosis-only* baseline approaches chance, at 0.562, and it is that near-chance baseline which manufactures the cohort's largest apparent effect, $\Delta\text{AUC} = +0.217$ with $p = 0.121$ and an interval $[-0.030, +0.485]$ that spans half the available range; a large ratio over a baseline that predicts almost nothing, on eight events, is not a signal.

The null is not explained by the validity radius: the *exact* per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error and is immune to the ρ argument, also adds nothing over the competitor set. Our reading is that SFI's information is already present in standard connectivity features (λ_2-alone, min-cut, percolation), but an incremental null does not by itself separate that redundancy from a feature that carries little predictive information at all. We separate the two in Section 4.3, directly and without reference to the label, where the redundancy account holds for the region-level aggregates and fails for the extreme and edge-level columns. Either way the incremental null is itself a label-dependent finding, separate from ρ.

Fibrosis-heterogeneity baseline comparison, and its withdrawal. This comparison is not the pre-registered secondary endpoint, which is the localization test reported in Section 4.10, but a further feature-set contrast pre-specified alongside the primary one. An earlier version of this analysis reported that SFI beat a fibrosis-heterogeneity-only baseline on the combined real cohort (GBT $\Delta\text{AUC} = +0.103$, $p = 0.009$) and read that as evidence that SFI carries connectivity information beyond the substrate rather than re-encoding fibrosis. That result does not survive, and we withdraw both it and the claim it supported. Two changes separate the old number from the new one, and they weigh unequally; both figures below are gradient-boosting ΔAUC against the same fibrosis-heterogeneity baseline, tested with the paired DeLong statistic. Correcting the UW substrate while holding the cohort at its original 144 subjects leaves the estimate largely intact ($+0.093$, $p = 0.013$), so the old result was not principally a substrate artefact; it collapses instead when the Roney arm is extended from 62 to the full 100 released meshes, to $\Delta\text{AUC} = +0.017$ ($p = 0.452$). An effect that disappears when 38 further real subjects are added was a small-sample fluctuation, and it was one of $\sim$30 DeLong comparisons besides. On the full cohort no comparison, primary or secondary, shows SFI adding information beyond either baseline.

The failure mode of the discarded estimate is worth recording. Under the sealed substrate the UW arm alone returned $\Delta\text{AUC} = +0.340$ ($p = 0.026$) on six positives, a value no one should have believed and which we did not flag at the time, and it was that arm which carried the

combined figure; a large point estimate resting on six events is a warning, not a finding.

Naive (ungrouped) comparison. The pre-registration asks for the naive ΔAUC beside the grouped one so that leakage is visible, and it is reported here for every real-cohort arm: Roney +0.017 (lr) and −0.011 (gbt), UW +0.066 and −0.160, combined −0.011 and +0.010, against the grouped values of Table 1. The gap has no consistent sign: the naive estimate runs above the grouped one in two of the six arms, both logistic regression on a single cohort, and below it in the other four including the pooled logistic arm, so nothing here reproduces the optimism ungrouped cross-validation is expected to show. That is expected and offers no reassurance: as the Methods note, patient-level grouping is close to a no-op on real anatomy, so these six numbers are not a leakage demonstration, and the demonstration instead rests on the replicated synthetic cohort, where resampled networks share a parent shape family.

Power analysis. In the synthetic cohort the subspace effect (+0.0028) needs ∼4,150 cases for 80% power and the single-vector effect (+0.0007) ∼23,000, clinically undetectable *there*. We do *not* transport "clinically undetectable" to the real cohorts, which are a separate regime. On the corrected substrate their estimates are +0.009 (Roney, $n = 100$) and +0.012 (combined, $n = 182$), with intervals that contain zero and exclude the 0.05 gate. An earlier version read the then-current +0.036–0.051 as compatible with a detectable effect we were underpowered to confirm; those estimates are superseded and that reading no longer holds.

4.2 The clinical endpoint, and the false positive its own protocol caught

Source: `results/clinical_endpoint.json`.

Every endpoint above is scored against a simulator's verdict, whereas this one is scored against patient outcomes. The cohort's authors supplied per-patient two-year post-ablation outcomes for all 82 UW patients (recurrence being at least 30 s of documented atrial fibrillation or flutter after a 90-day blanking period, with every patient followed for the full two years), and §8 of the pre-registration, which governs this analysis, was frozen and git-tagged before any feature was placed beside that column. The analysis was run once, and its result is reported as it came out.

Three commitments in that protocol, all of them predating the data join, govern how the result should be read. First, the 0.05 effect-size gate used for the simulator endpoint is *not* reused here: at $n = 82$ with 48 events, power at $\Delta\text{AUC} = 0.05$ is 0.24, so that gate would be unfalsifiable, and the minimum detectable effect was fixed at $\Delta\text{AUC} \approx 0.10$–$0.15$. Second, testing follows a fixed sequence and halts at the first non-rejection. Third, and decisively here, §8.6 anticipated that UW fibrosis is categorical, not continuous, and warned in writing that this "biases the comparison toward SFI, because a degraded baseline is easier to beat" and pre-committed that if our baseline came out materially weaker than the source study's published performance, any ΔAUC would be reported as *confounded by baseline degradation* and not as a win for SFI.

Under logistic regression the competitor baseline gives an out-of-fold AUC of 0.250, adding SFI gives 0.447, and the increment is $\Delta\text{AUC} = +0.197$ with a bootstrap interval $[+0.054, +0.334]$ and a paired DeLong $p = 0.0061$, clearing both pre-registered thresholds on real patients with a p-value an order of magnitude below 0.05.

The increment is nonetheless an artefact, and the pre-registered protocol classified it as one before we reached that judgement ourselves. A baseline AUC of 0.250 is anti-predictive, not weak; adding SFI reaches only 0.447, still below chance. Three checks confirm the reading: every one of the fifteen features, competitor and SFI alike, is univariately at chance on this outcome (0.435–0.547); the gradient-boosting arm, whose baseline sits at chance (0.516), gives $\Delta\text{AUC} = +0.021$ at $p = 0.738$; and when the nine SFI columns are replaced by nine columns

of standard Gaussian noise and the identical pipeline is re-run, the increment reproduces, the noise gaining +0.137 on average across 20 draws, with 6 of 20 matching or exceeding SFI's +0.197 (empirical $p = 0.333$). The logistic arm therefore measures dilution instead of prediction: padding an over-fitted, anti-predictive baseline with any nine columns pulls its AUC back toward 0.5.

The inversion itself follows from a property of the estimator rather than from anything about these patients. The folds carry very unequal event rates (25% to 81%), and because each training set is what remains once its test fold is removed, the two prevalences move in opposition. The model calibrates to its *training*-fold prevalence: the mean predicted probability on a held-out fold tracks that fold's *training* prevalence at $r = +0.995$, so the fold with the fewest events receives the highest predicted probabilities. Pooling out-of-fold probabilities across such folds inverts the ranking. Scoring within folds and averaging removes most of it ($0.250 \rightarrow 0.345$ for the baseline), which is a diagnostic and not a re-analysis.

The clinical primary endpoint is **not met**, and per §8.6 the observed increment is reported as confounded by baseline degradation. Under the §8.7 stopping rule the ablation-induced ΔSFI test is therefore not performed; it was not computed, since running it and then declining to report it would spend the very α the sequence protects. Two analyses outside the confirmatory family are reported for completeness: on the data provider's own 67/15 split (explicitly *not* independent, since those 15 patients also enter the primary, and not powered, with 8 events) the two classifiers disagree in sign (+0.196 and −0.107); and among the 48 recurrers, SFI does not separate the 35 who presented with fibrillation from the 13 who presented with flutter ($\Delta\text{AUC} = -0.009$ and -0.044), the expected direction given that flutter is typically macro-reentrant and often right-atrial.

We report this at length because it is the paper's thesis instantiated on real clinical outcomes rather than on simulated ones. A naive analysis (a $p < 0.05$ threshold, no effect-size gate, no baseline-quality check, no noise control) would have published $\Delta\text{AUC} = +0.197$ at $p = 0.0061$ as evidence that a spectral biomarker predicts atrial-fibrillation recurrence in patients. The guard that caught it was written into §8.6 on the day the data arrived, before any feature met the outcome column, by an author who expected to be the one fooled.

4.3 Testing the redundancy explanation directly

Source: `results/feature_redundancy.json`.

The account above attributes the null to feature redundancy, an explanation inferred from the incremental AUC being zero. A zero increment does not establish it, since a feature that is pure noise also adds nothing, as does one the classifier cannot exploit, and those are three different diagnoses. We therefore measured redundancy directly and without reference to the label, by regressing each SFI column on the competitor block and by taking the leading canonical correlation between the two blocks, with a permuted-SFI control to separate genuine redundancy from a degenerate estimator.

The blocks are almost collinear in their dominant direction: the leading canonical correlation between the nine SFI columns and the six competitors is 0.996, against 0.289 for the permuted control. Taken alone that supports the redundancy account, but the per-column picture is less uniform:

SFI column	R^2 on competitors	R^2 permuted
`sfi_total`, `sfi_region_mean`	0.970	0.008
`sfi_top3`	0.842	0.022
`sfi_top2`	0.665	0.040
`sfi_edge_frag_max`	0.205	0.010
`sfi_edge_frag_mean`	0.154	0.015
`sfi_region_max`, `sfi_top1`	0.138	0.007
`sfi_region_std`	0.098	0.006

Four of the nine columns are reconstructible from the competitors at $R^2 \geq 0.60$, and the two aggregate summaries almost perfectly so. The remaining five keep most of their variance, at $R^2 < 0.25$, well above the permuted control but far from redundant. The split does not fall along the aggregate versus order-statistic line: the competitors reconstruct the two region-level aggregates together with the deeper region order statistics `sfi_top2` and `sfi_top3`, while the edge-level pair, the region maximum, and the region spread all resist reconstruction.

This qualifies the null rather than overturning it, and sharpens what the null means. For `sfi_total` and `sfi_region_mean` the competitors already carry the information, and adding those two columns cannot help. For `sfi_region_max`, `sfi_top1`, and `sfi_region_std` there is genuine unique variance in the feature, and the finding is the stronger one: that variance exists and does not predict the label. A biomarker can fail either by saying nothing new or by saying something new that does not matter, and on this evidence SFI does both in different components, of which we had previously reported only the first.

4.4 Region definition and the patient-level reduction

The per-region SFI of Eq. (4) is stated over a region R, and until now we had left R implicit. Since the partition and the reduction that follows it produce the feature block whose incremental ΔAUC is the primary endpoint, both are given explicitly here. Neither was pre-registered (the pre-registration refers to the per-region SFI throughout without fixing a partition or a patient-level reduction), so both are logged deviations, fixed post hoc but without reference to the outcome, since SFI is closed-form given the mesh and is never fitted to the label.

A further deviation bears directly on the validity argument: every real-cohort number in this paper is computed from the first-order per-region SFI (`asb.sfi.sfi_region`). The pre-registration also specifies a gap-triggered switch to the subspace variant when the validity radius is exceeded, and that switch is not wired into the real-cohort feature path; the subspace estimator is used only in the synthetic validity experiments of Section 2.4. Since real atria sit at median $\rho \approx 2422$, the switch would have fired on every mesh had it been applied. We report the first-order numbers rather than silently re-deriving the table, since the ranking a classifier consumes is invariant to the global Δw magnitude that drives ρ, and since the question the switch exists to answer is settled directly in Section 4.1, where the *exact* per-region $\Delta\lambda_2$, which carries no first-order error at any $\|\Delta L\|$, also adds nothing over the competitor set. The reader should nonetheless know that the primary endpoint was scored with the estimator that the validity argument indicts.

Partition. Regions are a 3×3 grid over the universal atrial coordinates. With $(\alpha, \beta) \in [0,1]^2$ per vertex, a vertex is assigned $r = 3\lfloor 3\beta \rfloor + \lfloor 3\alpha \rfloor \in \{0, \ldots, 8\}$, so every mesh carries exactly nine regions regardless of its vertex count. An edge is assigned region r only when both endpoints carry r; edges straddling two regions go to a tenth bucket, $r = -1$, which is retained rather than discarded so that the per-edge sensitivities sum to the global total. The rule is shared by the closed-form and Monte-Carlo estimators so their region keys coincide.

At this step the two cohorts differ in kind and not only in degree: the Roney meshes carry anatomically registered universal atrial coordinates, so their nine regions are anatomical, whereas the UW release ships none, and we substitute the top two principal axes of the surface coordinates, min-max normalised. That surrogate is a smooth two-dimensional chart adequate for binning, but it is not registered to anatomy, and any region-level or localization claim on the UW cohort is correspondingly weaker. The substitution is recorded in the mesh metadata.

Reduction to the patient. The classification target is one binary verdict per patient, so the nine per-region values must be reduced. Nine scalars enter the design matrix, all in the `sfi_` namespace: the global `total` (the sum of expected per-edge λ_2 drop over all edges);

`region_max`, `region_mean` and `region_std` over the region values; the three largest region values in descending order, `top1`–`top3`; and `edge_frag_max` and `edge_frag_mean` over the raw per-edge fragilities $(\varphi_{2,i} - \varphi_{2,j})^2$. The order statistics are included because a fragility biomarker is a claim about the *worst* region rather than the average one, and reporting only a mean would have made the hypothesis untestable in its own terms. Two of the nine names denote one quantity: `region_max` and `top1` are both the maximum over the region values, and are identical on all 182 subjects to zero difference, so the block carries eight distinct columns under nine names and the two share a row of the table in Section 4.3.

Because the competitor features are patient-level scalars already, no analogous reduction is applied to them, and the comparison is between an add-on of eight distinct columns and a six-column baseline, which if anything favours the add-on. Nor is any of the nine columns fitted to the outcome: SFI is closed-form given the mesh and the frozen uncoupling field, so it contributes no tuning degrees of freedom, and the null cannot be attributed to a failure of feature selection.

4.5 A provenance audit of the released substrates, and four controls

Sources, all under `results/`: the cohort audit is `cohort_geometry_audit.json`, and the four controls are `uw_substrate_ablation.json`, `roney_fibre_control.json`, `roney_fibrosis_quantization.json` and `fibrosis_burden_swap.json`. Every Roney inducibility arm in this section runs on the 62-mesh subset that predates the extension of the predictive cohort to all 100 released meshes, and was not re-run on the full deposit; the predictive results of Section 4.1 use all 100.

This section is prompted by a discrepancy between the two cohorts under a protocol frozen before any label was generated and identical in every parameter: the Roney meshes returned 20/62 subjects inducible (32.3%) while the UW meshes returned 6/82 (7.3%) on the substrate as released. Three UW rates recur below and are kept distinct throughout: 6/82 as released, 8/82 after the cap repair (the substrate every predictive result uses), and 14/82 after both repairs. A fourfold discrepancy on cohorts we had assumed carried comparable fibrosis burden, an assumption Control 4 below overturns, should be treated as a hypothesis about the pipeline rather than accepted as a property of the cohorts. We had previously recorded the UW arm as uninformative and moved on, and that was the error. The audit is set out first, followed by three controls that do not explain the anomaly and a fourth that does.

Two problems sit between the release and our use of it, and neither is a defect in the data; both follow from an undocumented release being read by someone who guessed. The first guess is ours: the only per-element scalar the UW release carries is `elemTag`, the Dryad README does not define its values, and we had been mapping tag 164 to half-fibrotic tissue. Four measurements show that it is not tissue at all. It resolves into four to six large connected components in every one of the 82 meshes, five in 75 of them, four in five and six in two, which is the ordinary variation of pulmonary-vein anatomy: a left common trunk gives four, a right middle vein gives six. Each large component is topologically a disc, 11–42 mm across, sitting at 0.47–0.82 of the atrial radius from the centroid. It borders fibrotic elements on about 0.08% of its incident edges against a chance rate of 14–20%, a depletion beyond a hundredfold that is incompatible with reading it as a fibrosis border zone. It is also ablation-invariant, holding at 20.0% of elements before the procedure and 19.9% after, while every other class moves. Tag 164 is the set of caps sealing the pulmonary veins and the mitral valve. Removing it opens the surface: on 80 of the 82 meshes the Euler characteristic falls, by a median of 7 (range 2–12). We report the *change*, not an opening count, because $\chi = 2 - k$ presumes a clean closed surface and these do not start as one: χ before removal spans -18 to 5. The two exceptions differ in the direction of the change, not only in their starting value: on ID040 removal takes χ from -16 to -8 and on ID049 from -18 to -5, so the full-cohort range of the change is -13 to 12 (`results/uw_euler_census.json`). We state the split because the census in this paragraph replaces an

earlier claim generalised from one mesh, and quoting the 80-mesh figure over 82 would repeat that error in the correction itself.

Treating those caps as half-conducting tissue inflates fibrosis, and it also seals the atrium's orifices, so that simulated activation crosses the mitral valve and the vein ostia instead of circling them, and reentry anchored on those orifices, a principal mechanism of atrial fibrillation, cannot form.

That identification was made from geometry alone, before any legend for the tags existed, and has since been confirmed by the cohort's authors, who define 164 as veins and valves and treat them, as we now do, as electrically dead. The same correspondence settles a tag this study never encounters: 199 is ablation scar and is likewise dead, where our mapping had it as maximally fibrotic *but still conducting*. It occurs only in the post-ablation meshes and a census confirms it is absent from all 82 pre-ablation meshes, so no result here is affected; the code now refuses to load any mesh carrying a dead tag it has not been told to remove, so the assumption cannot persist unchecked. We record the confirmation because a census that agrees with the ground truth is worth more as evidence than the same census standing alone, and because the audit below rests on it.

The second problem is upstream, and it is a placeholder mistaken for data. A `VECTORS fiber` array is present in all 164 released meshes, but it equals the constant $(1, 0, 0)$: the measured mean directional spread is exactly 0.000. The Roney cohort, by contrast, spans 0.922–0.987 across its 62 meshes (mean 0.954), with roughly one distinct direction per vertex. Conduction anisotropy is applied relative to the local fibre direction, so a single global direction reduces it to a fixed coordinate bias carrying no anatomy. The same cardinality argument applies to fibrosis: three distinct levels in the UW release, effectively two once the caps are dropped, against 700–1646 (mean 1318) in Roney at the same 2000-node coarsening.

Here too the cohort's authors have since supplied the explanation, and it is the benign one: fibre orientations are withheld under the same IRB as the intensity ratios, and are meant to be regenerated with published tools, not read off the release. The constant array is a placeholder, and nothing is wrong with the deposit; what was wrong was our reading it as anatomy, the same failure as the tag mapping and with the same root in an undocumented file used without asking. Both were caught by measurement rather than by correspondence (the degeneracy is visible in the array itself), but neither would have occurred had we asked first.

Control 1: repairing the substrate. Opening the orifices and supplying a spatially varying fibre field, in a 2×2 design over all 82 subjects (Fig. 2a), raises inducibility from 6/82 to 14/82. Either repair alone converts two subjects ($6 \to 8$ in both single-factor arms) and together they convert eight, so on these four cells the averaged main effect of a repair and the interaction contrast are both +4 subjects and cannot be told apart; we do not claim the two act super-additively. The mechanistic reading an interaction would invite (that reentry needs both an anatomical obstacle and anisotropic heterogeneity) is in any case undercut by Control 2 below, which shows the fibre factor does not move the rate at all on a cohort that has a real fibre field. We do not present this as a fix: the strongest contrast does not reach paired significance (McNemar exact $p = 0.077$), it closes only 39% of the distance to Roney, and only two of the original six positives survive the repair. The varying fibre field is itself a probe rather than a correction: it is the global $+z$ direction projected onto each vertex tangent plane, and it is not anatomy.

Control 2: destroying the fibres on a cohort that has them. The cleaner experiment replaces the Roney fibre field with the same constant $(1, 0, 0)$, holding anatomy, fibrosis and every solver parameter fixed (Fig. 2b). The rate does not move at all: 20/62 with real fibres and 20/62 without, McNemar $b = c = 5$, $p = 1.0$. Fibre degeneracy does not drive the inducibility rate, and our first attribution of the UW anomaly to it was wrong. The perturbation is real

at the level of the dynamics (sustained reentry duration changes on 50/62 subjects, by up to 661 ms, and 10/62 verdicts flip), but those effects cancel in aggregate.

Control 3: destroying the fibrosis gradation. The remaining structural difference was that UW fibrosis is binary where Roney's is graded, and graded border zones are where slow conduction and unidirectional block are supposed to arise. Binarising the Roney fibrosis field should therefore have suppressed reentry, and it does the opposite (Fig. 2c). Thresholding at $f > 0.5$ raises inducibility from 20/62 to 31/62; thresholding at the per-subject quantile that holds mean fibrosis burden fixed at 0.333, so that only the gradation is destroyed, raises it to 40/62. Coarsening the fibrosis field roughly doubles inducibility. In hindsight the mechanism is unmysterious (a sharp binary boundary is a stronger source–sink discontinuity than a smooth ramp), but the direction is the opposite of our prediction, and it disqualifies coarse fibrosis as an explanation for a cohort that is inducible *too rarely*.

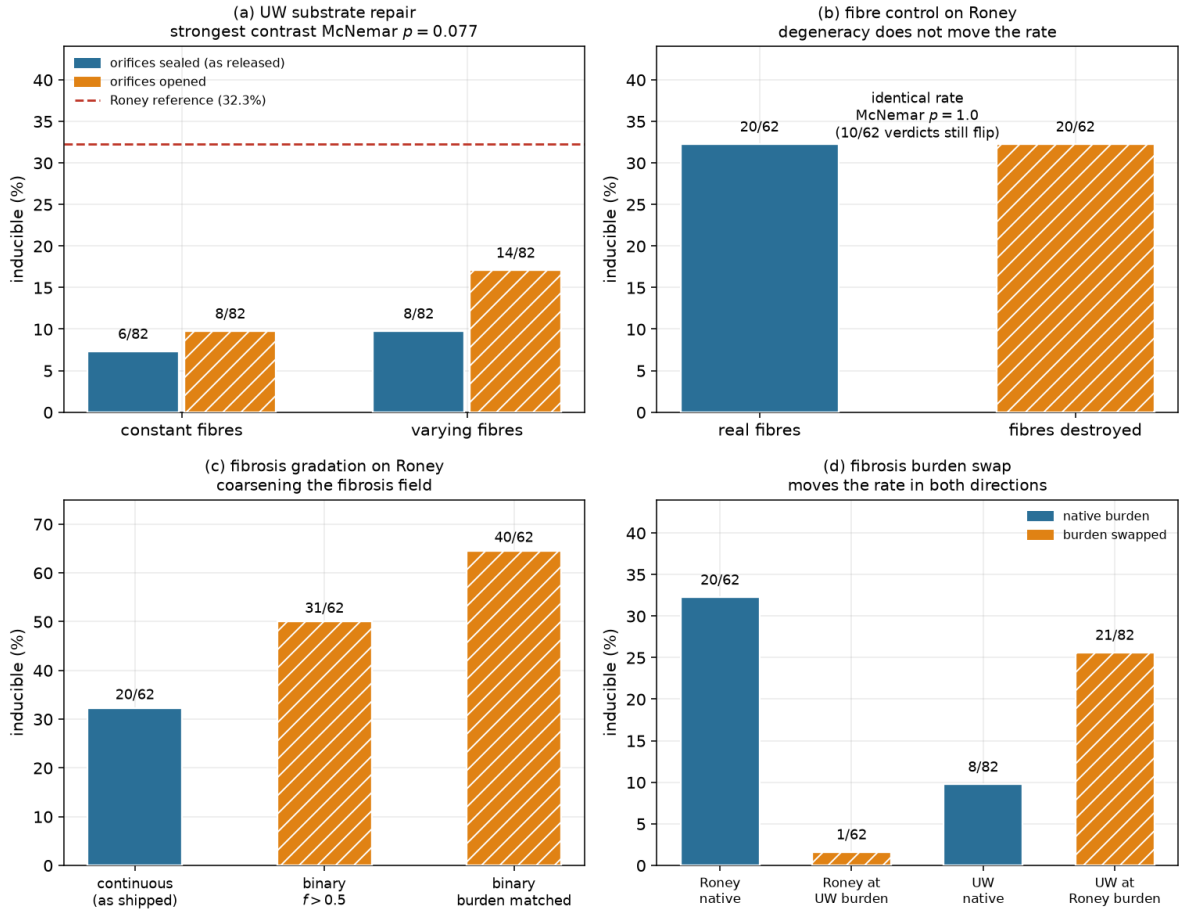


Figure 2: Four controls on the substrate. (a) Repairing both UW defects raises inducibility but not significantly, and not to the reference level. (b) Destroying the fibre field on a cohort that has one leaves the rate exactly unchanged. (c) Destroying the fibrosis gradation while holding mean burden fixed roughly doubles the rate, the opposite of the predicted direction. Panels (a)–(c) therefore leave all three substrate hypotheses unsupported, which is weaker than excluding them, and none of the three accounts for the anomaly. (d) Swapping fibrosis burden between the cohorts moves the rate in both directions and resolves the anomaly: Roney at UW's burden collapses to 1/62, UW at Roney's rises to 21/82. The dashed line is the Roney reference, 32.3%.

Taken together the three substrate controls leave the anomaly unexplained, and they tighten rather than loosen it: of the two defects we can manipulate, one does not move the rate at all

and the other, imposed deliberately, moves it upward. None of the three establishes an effect, which is weaker than excluding one, and none of them points at a substrate defect as the cause. The search accordingly moved from what might be broken in these meshes to what they contain.

Control 4: the cohorts do not carry comparable fibrosis. Having exhausted the defects, we measured what the solver is given. With the meshes coarsened to the same 2000 nodes, mean fibrosis over the meshes these controls simulate is 0.333 on Roney's 62 against 0.248 on UW's 82: the UW cohort carries about 25% less. (A separate 40-mesh-per-cohort geometry audit gives 0.349 against 0.244, with the fraction of tissue above half-fibrotic at 0.310 against 0.239; we quote the simulated meshes here because they produced the rates.) That deficit is measured on the cap-repaired substrate, which matters for reading the history. As released, with the caps still counted as half-fibrotic, UW measured 0.307 against Roney's 0.333 (only about 8% lower), so the cap defect was itself masking most of the burden difference, and the 6/82 that opened this section was produced under that masking. Every predictive result in the paper uses the repaired substrate, where the deficit is the full 25%. A second qualification limits how the deficit should be read: the two burdens are not measured the same way. Roney's is the mean of a continuous IIR-derived ramp; UW's, whose intensity ratios are withheld, is the mean of a two-valued tag classification, so it is closer to a fibrotic-area fraction than to a mean severity. Both enter the solver through the identical conductance ramp, so the comparison is operationally sound (the solver sees the same kind of field either way), but 25% is a difference between two differently constructed quantities and should not be read as a claim that these patients carry a quarter less fibrosis in any imaging sense. What does not depend on the comparison is the swap below, which rescales each cohort's own field and is therefore internal to it. The frozen protocol's own calibration reports a rank correlation of 0.75 between fibrosis and sustained reentry, so a cohort with less of it should be less inducible, and the lower rate is therefore the labeller behaving as specified rather than an anomaly.

Swapping the burden between cohorts confirms it in both directions (Fig. 2d). Rescaling Roney's fibrosis multiplicatively to UW's level, which preserves the spatial pattern and the gradation and moves only the level, collapses inducibility from 20/62 to 1/62 (Fisher exact $p = 4.0\times10^{-6}$). Rescaling UW's upward toward Roney's raises it from 8/82 to 21/82 ($p = 0.013$). Both control arms reproduce their native rates exactly. The effect is large, symmetric, and an order of magnitude better supported than any of the three substrate hypotheses.

At matched burden the point estimates invert, which bears on the original question: Roney rescaled to 0.241 gives 1/62 while UW at 0.248 gives 8/82. On one and eight events that contrast does not establish a reversal and we do not claim one; it shows only that the UW cohort is not intrinsically resistant to reentry. Most of the apparent anomaly is accounted for by a difference in how much fibrosis the two released datasets contain, and the two released defects, real as they are, were never the explanation; the burden account does not cover openCARP's UW rate, which the same rescaling leaves at 0/30 (Section 4.6). We had spent three controls looking for something broken before checking whether the two cohorts were comparable in the first place: an out-of-range aggregate is a hypothesis about the pipeline, but a hypothesis about the inputs first.

4.6 An independent solver as a robustness check

Sources, all under `results/`: `opencarp_full_cohort.json`, with the per-cohort and chance-corrected concordance in `opencarp_concordance_detail.json` and the endpoint re-scored against openCARP's verdicts in `gm1_opencarp_labels.json`; all three calibration attempts, including the two rejected, are in `opencarp_calibration_history.json`, with the discretisation and propagation checks in `opencarp_bin_sensitivity.json` and `opencarp_uw_propagation.json`.

Every in-silico result above rests on one phenomenological labeller, a monodomain Mitchell–Schaeffer solver substituted for the pre-registered openCARP ground truth when openCARP could not be installed. That substitution is now testable, because openCARP runs. Replacing the labeller with it would invalidate every number at once, so we instead run both solvers and report their agreement.

Calibration. openCARP ships the same Mitchell–Schaeffer membrane model, so the comparison is between solvers rather than between physics, but its defaults are not ours. Two quantities were matched to the frozen configuration, action-potential duration and conduction velocity, each measured independently of the endpoint. For action-potential duration, openCARP's default `tau_close` 150 gives APD90 = 277 ms, against the frozen 218 ms healthy and 121 ms fibrotic; measured on this build, `tau_close` 110 $\rightarrow$ 209 ms and 55 $\rightarrow$ 113 ms, so $115 \times (1 - 0.5f)$ reproduces both within about 2%. For conduction velocity, the shipped conductivity gives 0.344 m/s against the frozen band of 0.4–1.2 m/s. That is a wavelength of roughly 75 mm in an atrium of $\sim$110 mm characteristic length, and it makes reentry far too easy. Raising the intracellular conductivity alone saturates, because the monodomain conductivity is the harmonic mean of the intra- and extracellular values; scaling both gives 0.852 m/s at $g_{il} = 1.00$, and we adopt 1.05.

Two configurations were rejected before this one, at 90% and 76.7% inducible with *negative* fibrosis–verdict correlations, the signature of rate-dependent block and not of reentry. We record that because it is the protocol working: no endpoint parameter was adjusted at any stage, so the calibration gate remained able to fail, and on those two configurations it did.

On the calibrated configuration the pre-registered gate is met on Roney but not on UW. Roney gives 22/100 inducible (22.0%, inside the 10–40% band) with a fibrosis–verdict rank correlation of +0.404, against the frozen solver's 0.75, whereas on UW openCARP returns 4/82 inducible (4.9%, below the band) with a rank correlation of only +0.079. Across all 182 subjects the two solvers agree on 156, or 85.7%. That figure is raw and has not been corrected for chance. On an outcome this imbalanced the correction matters: Cohen's κ is 0.536, and the two solvers agree on only 21 positives. The cohort-level split shows why the pooled number should not be read as a tissue-level result. On Roney, where the gate is met, raw agreement is 82/100 (82.0%) at $\kappa = 0.561$; on UW, where it is not, raw agreement is higher at 74/82 (90.2%) but κ falls to 0.287, because at a 4.9% and 9.8% event rate almost all of that agreement is shared negatives. The pooled figure is therefore a statement about the protocol across both cohorts, not about the tissue, and the arm that looks most concordant is the one carrying the least information. The substantive part is the asymmetry: of the 26 subjects the two solvers disagree on, openCARP calls only 5 inducible where the monodomain solver does not, while the reverse happens 21 times; over the whole cohort openCARP labels 26/182 against monodomain's 42/182. Sampling noise does not explain that imbalance (McNemar exact $p = 0.0025$); it is systematic, so the two solvers differ in level and not only in which subjects they pick. Relative to an independent reaction–diffusion solver, then, the stand-in is the more liberal of the two and does not undercall, so the incremental comparisons reported above are conservative rather than inflated. (On the 60-subject calibration subset the discordance was entirely one-way, 0 against 5; that does not hold on the full cohort and the figure quoted here is the full-cohort one.)

The comparison re-scored against openCARP (exploratory). Agreement on the label does not by itself bound the incremental ΔAUC, so we ran the comparison instead of inferring it: the identical GM1 evaluation, on identical features, with the outcome column swapped from the monodomain verdict to openCARP's (`results/gm1_opencarp_labels.json`). This is *not* an endpoint and cannot become one. The pre-registration binds openCARP labels out of any endpoint until the calibration gate is re-passed on that solver, and the gate failed on UW; the arm is reported as an exploratory sensitivity analysis under that precondition, and no

gate verdict is claimed from it. The null holds: the combined competitor-baseline increment is $\Delta\text{AUC} = +0.028$ ($p = 0.328$), against $+0.012$ under the stand-in; per cohort it is $+0.033$ on Roney ($p = 0.211$) and $+0.067$ on UW ($p = 0.445$), and no arm in either family clears the pre-registered 0.05 gate at $p < 0.05$. This arm is less powered by construction, since openCARP calls 26 subjects inducible against the stand-in's 42 and UW contributes only 4, so the intervals are wider and the combined one, $[-0.027, +0.085]$, no longer excludes the gate the way the stand-in's does; the UW arm here also sits on the cohort whose calibration gate failed. The result is therefore that the null is not an artefact of the labeller, not that openCARP independently establishes it.

The check is bounded in two ways. The fibrosis field must be discretised into conductivity classes for openCARP, and the verdict is unstable at five classes (1/20) though identical at ten, twenty and forty (7/20, with complete per-subject agreement), so the operating choice of twenty sits inside the converged regime while the coarse limit is not benign. The larger bound is openCARP's 4/82 on the UW cohort, which unlike the monodomain solver is *not* recovered by raising that cohort's fibrosis burden: rescaling UW upward leaves the rate at 0/30. The mundane explanations do not survive: propagation failure is ruled out, because on UW meshes the wave reaches 99.4% of nodes and peak depolarisation averages 0.83, so activation spreads across the surface and then dies; it is not confined to the stimulus site. The coarser discretisation is ruled out as well: at the same node budget the 40-mesh-per-cohort geometry audit gives UW a mean edge of 3.12 mm against Roney's 2.91 mm, but the convergence study shows coarser meshes *over*-call (58.3% at 1500 nodes), so that difference predicts a higher UW rate, not a near-zero one.

What remains is a genuine substrate difference that the burden rescaling does not capture, and we do not know what it is. The monodomain burden result (Section 4.5) is established in both directions on that solver and is unaffected by this; what is not established is that openCARP's near-zero UW rate has the same cause. We report the discrepancy rather than resolve it, and restrict the concordance claim to the cohort where both solvers are calibrated. A robustness check that agreed everywhere would be less informative than one that disagrees somewhere and reports the disagreement.

4.7 The labeller is reproducible in aggregate and unstable per subject

The controls expose a property of the ground truth that matters more than the substrate defects do. Destroying the Roney fibre field holds the aggregate rate at exactly 20/62 while flipping 10/62 individual verdicts; repairing the UW substrate retains only two of its six original positives; and the six-tier convergence study reported below finds only 7/24 subjects consistent across resolutions. The aggregate inducibility rate is therefore a stable quantity, while the per-subject verdict is not.

The consequence falls on the endpoint itself: the pre-registered comparison is scored on per-subject labels, and non-differential noise in a binary outcome attenuates any incremental AUC toward zero. Part of the null we report is thus a property of the labeller rather than of the biomarker, and we cannot cleanly separate the two with the present ground truth. The same noise sets a floor on what any feature could achieve here, a limit that applies to the whole class of studies scoring spectral or structural features against a phenomenological inducibility verdict rather than to SFI alone. The pre-registered openCARP ground truth, substituted here by a monodomain Mitchell–Schaeffer solver, would need to be run before the residual is attributable.

4.8 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$, while real atria sit at median $\rho \approx 2422$. This proves the single-vector estimator is numerically invalid on real tissue and φ_2 is ill-conditioned, but (as stated in Section 2.4) it does not by itself prove

non-predictiveness; the predictive null of Section 4.1 is a separate, label-dependent finding, and Section 4.3 shows redundancy accounts for only part of it.

4.9 The ρ scaling law

The gap collapses with system size by Weyl's law [6]: for a d-dimensional domain the low Laplacian eigenvalues scale as $\lambda_k \sim (k/N)^{2/d}$, so ρ grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs. N at fixed mean degree, five-point fits). The atrial row is measured on the *weighted* conduction Laplacian the biomarker itself uses, on the corrected substrate (`results/rho_scaling_weighted.json`); the random-geometric rows are unweighted, those graphs carrying no conductances (`results/rho_scaling.json`).

medium	fitted gap exponent	reference
real atrial surface (2-manifold)	$N^{-1.00}$	Weyl 2-manifold -1.0 (within 0.5%)
2-D random-geometric	$N^{-1.81}$	faster than naive $2/d$
3-D random-geometric	$N^{-0.84}$	faster than naive $2/d$ (non-monotone series)

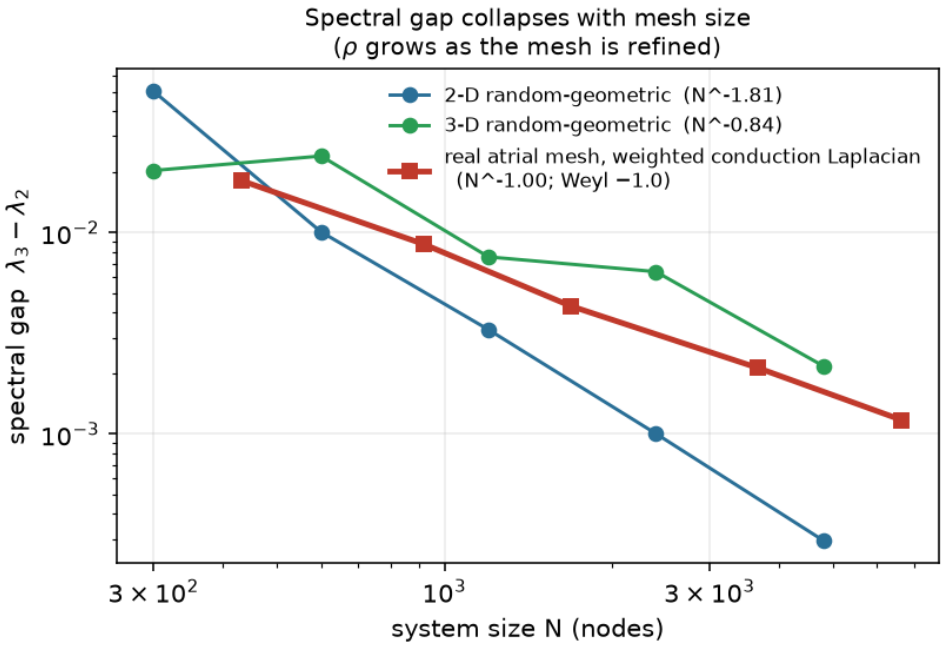


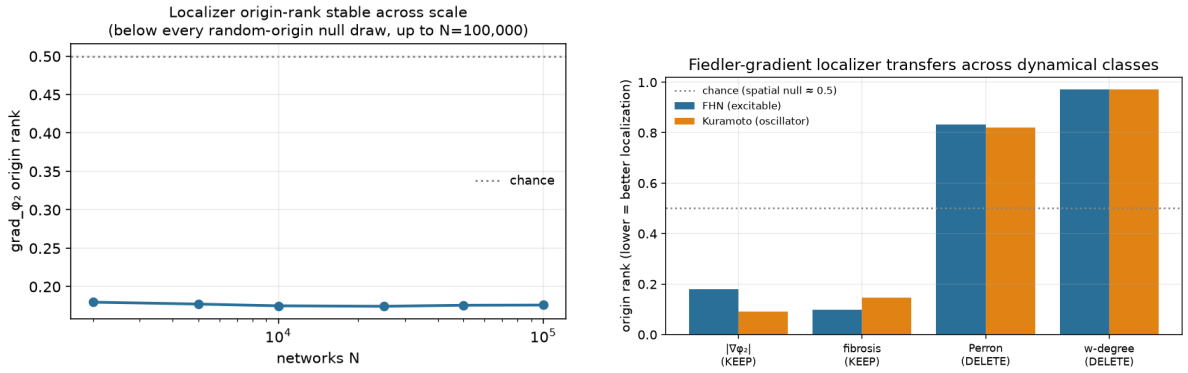
Figure 3: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer, the opposite of the expectation that "more resolution helps." Only the real atrial 2-manifold cleanly matches the Weyl exponent (-1.00 vs. -1.0); the random-geometric exponents are reported as "at least as fast as Weyl," not law confirmations, and are measured on the unweighted combinatorial Laplacian, those graphs carrying no conductances. The atrial arm is one released mesh resampled across five coarsenings, so it describes how the gap of a diffusively coupled 2-manifold scales with resolution rather than a property of any one patient. An earlier version measured this arm on the unweighted Laplacian and on the superseded substrate of Section 4.5, giving -1.05; correcting both moves it to -1.00, and the solver plays no part — shift-invert and the original path agree to every digit in every arm.

4.10 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including $|\nabla\varphi_2|$: its median *argmax-geodesic error* —the shortest-path distance along the surface from the localizer's argmax to the simulator's reentry origin, expressed as a fraction of the mesh's maximum geodesic extent rather than in millimetres — is 0.472 of the maximum geodesic extent (48.3 mm), with a bootstrap 95% CI of $[0.350, 0.656]$, and it is *not* below the spatial-null 5th percentile 0.406 (rotational/torus-shift permutation $p = 0.229$). Because the interval straddles that threshold, on the 20 localized subjects the endpoint is neither met nor resolved. The result that survives is weaker: $|\nabla\varphi_2|$ is a *rank* localizer (the true origin sits in its top $\sim$18%) that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media (cardiac monodomain arm: `results/gm2_metrics.json`; FHN arm: `results/gm4_100k_metrics.json`; Kuramoto arm: `results/gm4_kuramoto_metrics.json`). *Origin rank* is the fraction of nodes whose localizer value is at least the value at the true reentry origin, so 0 marks a perfect localizer and 0.5 chance; it is scale-free across media of different size. *Null draws* counts resampled random-origin draws whose mean origin-rank is at or below the observed one, the pooled null being resampled to the smallest per-subject sample: 300 draws in the cardiac arm and 200 in the FHN and Kuramoto arms. The count is zero in every arm, which bounds the permutation p below $1/300 \approx 0.003$ in the cardiac arm and $1/200$ in the other two, but reports the resolution of the draw count rather than the strength of the effect.

medium	$\|\nabla\varphi_2\|$ origin-rank	null draws $\leq$ obs.	fibrosis rank	n origins
FHN networks (100k)	0.176	0	$\sim$0.10	36,540
Kuramoto oscillators (500)	0.091	0	0.147	243
cardiac monodomain	$\sim$0.28	0	(co-localizes)	only $\sim$20



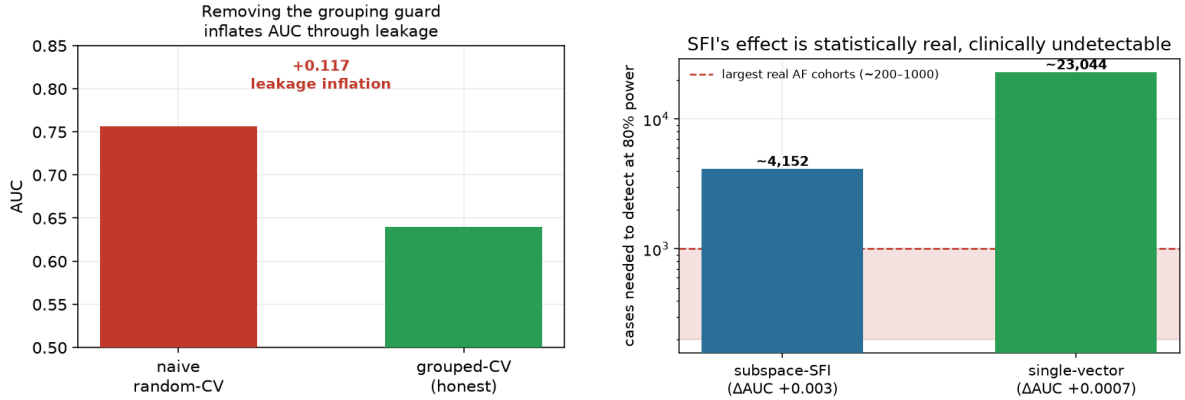
(a) origin-rank vs. scale

(b) transfer across media

Figure 4: The weak-rank localization pattern replicates across excitable and oscillator dynamics. The strict endpoint is nonetheless null, so $|\nabla\varphi_2|$ is *not* a precise localizer; the substrate field co-localizes, and a controlled dissociation experiment accounts for that co-localization (in a diffusively coupled medium the Fiedler bottleneck *is* the low-conduction lesion), so $|\nabla\varphi_2|$ matches substrate imaging rather than beating it, and substrate is the better localizer in the FHN arm; and the cardiac arm rests on only $\sim$20 origins.

4.11 The falsification protocol: what caught the false positive

Removing any one guard manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort), and the effect-size gate catches the p-value trap, in which subspace-SFI over the competitor set reaches a paired DeLong $p \approx 2 \times 10^{-24}$ at $N = 55{,}000$ while its $\Delta\text{AUC} = +0.003$ would need $\sim$4,150 cases to detect, so the difference is statistically significant and practically useless.



(a) leakage guard

(b) effect-size / power gate

Figure 5: A naive analyst (random CV, $p < 0.05$, no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker, whereas the pre-registered protocol correctly rejects it.

4.12 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion d_0 ($\mu^* = 500.7$), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out $R^2 = 0.35$ (large residual anatomical/label variance beyond the screened knobs, a label-noise caveat and not a positive). Across uncoupling strengths $\Delta w \in \{0.18, 0.36, 0.54\}$ the grouped GBT increment is $\Delta\text{AUC} = 0.033$–0.036 with all $p \geq 0.26$. This sweep runs on real anatomy rather than the synthetic media: it uses the GM1 cohort, and the original 62-mesh Roney subset that predates the extension to all 100. It therefore speaks to the real-cohort null and not to the synthetic one. Every arm sits below the pre-registered 0.05 gate, but non-significance on its own does not make these arms null, and because we report no interval for them they are indeterminate rather than a demonstration that the null is robust to Δw.

4.13 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans $\sim$5% to $\sim$85% depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller's Roney 34.0% (34/100) sits in the moderate-burst band (Marquardt et al. 2018 [15] report 30.6% in patients without prior AF), between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with $\sim$42% post-ablation (Liu et al. 2020 [18]). The UW 9.8% (8/82 on the cap-repaired substrate that every predictive result uses) sits just above the no-AF-control floor and far below the moderate-burst band, for a cohort that *is* AF patients; we flag that rate as anomalous, and Section 4.5 traces it to a

difference in fibrosis burden between the two releases rather than to biology or to the defects in the released UW substrate. The Limitations list two substrate defects, and repairing both moves the rate from 6/82 to 14/82, a difference that does not reach paired significance (McNemar exact $p = 0.077$, $n = 82$) and that retains only two of the six original positives; a direct control shows that the fibre defect is not responsible for any of it. This is a base-rate plausibility check and not clinical validation.

4.14 Resolution convergence: the labeller does not converge in the tested window

The convergence study reported here is the project's single weakest point, and we state that at the outset. On a completed six-tier study (Roney, $n = 24$ at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate falls sharply from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then appears non-monotone across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% at the finest affordable tier ($\sim$50k nodes), a 17-percentage-point spread over the last three tiers, so it has *not* settled.

At $n = 24$ per tier the precision is low, which bounds how much of that pattern can be treated as real. Exact binomial intervals are $[36.6, 77.9]$, $[4.7, 37.4]$, $[9.8, 46.7]$, $[7.1, 42.2]$, $[9.8, 46.7]$ and $[18.8, 59.4]$ percent across the six tiers. These are marginal intervals: the same 24 subjects are simulated at every tier, so each describes one tier's own precision and discards the pairing, and overlap between two of them is not a test of their difference. We use them only for the negative claim, namely that every tier from 3000 nodes upward overlaps every other, so the apparent wiggle among the fine tiers is *not* distinguishable from sampling noise and we do not claim it is a real non-monotonicity.

The paired comparison the intervals cannot make is available, because the same 24 subjects run at every tier: for each adjacent pair the 2×2 table is fixed by the two marginals and the agreement count, so an exact McNemar test follows without any further simulation (`results/convergence_paired_tests.json`). That test separates the two halves of the pattern cleanly. The coarse-mesh drop is real, with 11 subjects losing their verdict between 1500 and 3000 nodes against 1 that gains one, $p = 0.0063$, and it survives Holm correction across the five adjacent comparisons ($p = 0.032$); none of the four fine-tier transitions is resolvable ($p = 0.375$ to 1.000 unadjusted, all 1.000 after correction). What the study establishes is therefore sharper than the intervals alone allow and still sufficient for our purpose: the coarsest tier demonstrably overcalls, while across the fine tiers the rate cannot be shown to have converged and no tier can be declared the settled value. Distinguishing these tiers would require several hundred subjects per tier, roughly two orders of magnitude more simulation than the 144 runs here.

Table 4: Inducibility rate vs. mesh resolution (Roney, $n = 24$ per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement is itself non-monotone (50%, 75%, 71%, 63%, 79%) and the rate keeps drifting. At $n = 24$ per tier the exact binomial intervals overlap for every tier from 3000 nodes upward, so the ordering among the fine tiers is not resolvable; the claim supported is non-convergence, not a specific shape.

nodes	1,500	3,000	6,000	12,000	24,000	48,000
inducibility rate	58.3%	16.7%	25.0%	20.8%	25.0%	**37.5%**
adj.-tier agreement	–	50%	75%	71%	63%	79%

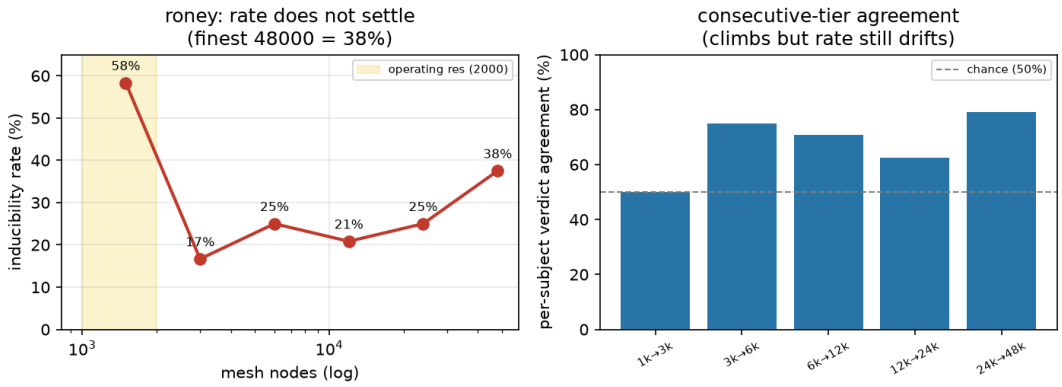


Figure 6: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement is non-monotone (50%, 75%, 71%, 63%, 79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

The convergence finding limits the absolute numbers but not the comparative claims. The rate does *not* collapse to zero and the phenomenon is real; what is resolution-sensitive up to $\sim$50k nodes is the magnitude. Because the main results ran at 2,000 nodes, every absolute number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. The two claims we lead with are untouched: (a) the relative SFI-vs-competitor comparison, since both arms score the *same* labels, so the shift in absolute AUC is common to them; and (b) the label-free estimator-validity argument (ρ is computed from the graph Laplacian, no labels at all). Sharing labels does not make the relative comparison unbiased: per-subject label noise attenuates the incremental ΔAUC toward zero (Section 4.7), so that comparison is conservative.

5 Discussion

The four contributions are (1) a pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker, yielding a clean null at synthetic scale (10^5 networks), where the mechanism remains inferred, and a measured null on 182 patients of real anatomy, where it is measured directly and label-free (Section 4.3); (2) a label-free validity criterion ρ for the single-vector estimator, which proves φ_2 is ill-conditioned on real tissue ($\rho \approx 2422 \gg \rho^* \approx 3$) and supplies a portable *a priori* check for any Fiedler-based marker, with the caveat that ρ bounds estimator accuracy, not predictive content; (3) a weak-but-real, purely connectivity-derived rank localizer that beats centrality baselines across three media but does not beat substrate localization, the better localizer in the FHN arm (Table 3), leaving the strict pre-registered localization endpoint null; and (4) a demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the p-value trap), with two worked instances from this project's own pipeline, one in silico at $p = 0.009$ and one on real patient outcomes at $p = 0.0061$ (Section 4.2), the second caught by a mitigation fixed in writing before the outcomes were joined.

5.1 Three corrections, and what caught each

Three substantive claims in earlier versions of this work did not survive, and because they are referred to individually where they arise, we set them out together here. This is not the complete list: the project notebook holds the full dated record, including further corrections that moved individual reported numbers.

The first correction moved two numbers at once, since both rested on the same cohort. An earlier version reported a combined real-cohort ΔAUC of $+0.051$ and read the primary endpoint as missed only on significance; separately, a pre-specified fibrosis-heterogeneity comparison reported $\Delta\text{AUC} = +0.103$ at $p = 0.009$, which we read as evidence that SFI is more than re-encoded fibrosis. Correcting the UW substrate at the original 144 subjects took the first to $+0.036$; extending the Roney arm from 62 to all 100 released meshes then took it to $+0.012$ and the second to $+0.017$ ($p = 0.452$). Both were caught by more data, and for the primary endpoint the larger share of the drop came from sample size rather than substrate. The withdrawn positive was vulnerable in the first place because it rested largely on one cohort's $+0.340$ over a near-chance baseline on six events.

The second correction concerns a causal attribution: we reported that the degenerate fibre field in the UW release was the likely reason that cohort is so rarely inducible. A direct control caught it, since destroying the fibre field on a cohort that has an intact one left the inducibility rate at exactly 20/62, $p = 1.0$. The explanation was plausible and mechanistically motivated, and it was nonetheless wrong; no amount of further argument would have revealed that.

The third correction concerns a set of geometric figures. We stated that element tag 164 formed exactly five components, had exactly zero adjacency to fibrosis, and left Euler characteristic -3 on removal; all three figures had in fact been generalised from a single mesh or from a rounded mean. They were caught by regression tests written over the whole cohort, two of which failed on first run. The identification of tag 164 itself survived and is better supported by the census than by the original figures.

The three share a common feature: each was caught by a control, a larger sample, or a test that this project designed and ran against its own results before submission, and none was found by a reader. The first correction is also a worked example of what the paper argues elsewhere, in that naive methodology (a $p < 0.05$ threshold with no effect-size gate, no multiplicity accounting and no replication on more data) would have reported $\Delta\text{AUC} = +0.103$ at $p = 0.009$ as a finding. It was produced by our own pipeline and it was false; the protocol that rejected it is the same one applied to the primary endpoint throughout.

The validity screen itself is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability, and the same precondition applies in each case: compute $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$; if $\rho \gg 1$, the first-order spectral marker is outside its validity radius, so its *numerical* $\Delta\lambda_2$ estimates and the stability of φ_2 cannot be trusted. This is a statement about the estimator and not about whether the feature ranks cases usefully, which must still be tested against competitors.

What this work does not claim should be equally clear. The mathematics is classical, and no new theorem is offered. ρ does *not* prove SFI is non-predictive, only that the single-vector estimator is ill-conditioned; the predictive null is separate from it, and is only partly redundancy (Section 4.3). The real-cohort result is a measured null in the sense that the interval excludes the pre-registered gate, but it does *not* exclude a small effect: the lower bound is -0.017, so an increment of a few thousandths remains compatible with the data and would need a far larger cohort to resolve. The localizer is *not* precise (strict endpoint null) and does not beat substrate imaging. The ρ scaling "law" is a clean match to Weyl only on the real atrial 2-manifold. Labels throughout this section are a monodomain simulator, not clinical outcomes and not openCARP; the one endpoint scored against real outcomes is Section 4.2, which is not met and whose apparent positive is an artefact.

6 Limitations

1. The labeller is strongly resolution-sensitive and does not converge in the tested window (Section 4.14): the rate drops from 58% at 1500 nodes to 17% at 3000 and rises to 37.5% at $\sim$50k nodes, above the 34.0% rate at the 2,000-node operating resolution, so the direction of

the residual bias there is not established. At $n = 24$ per tier the intervals overlap for every tier from 3000 nodes upward, so no ordering among the fine tiers is resolvable and we do not claim that the non-monotonicity there is real; only 7/24 subjects are fully consistent across all six tiers. All absolute rates and AUCs are coarse-mesh-provisional, while the relative comparison scores the same labels in both arms, so the shift is common to them, though per-subject label noise still attenuates the incremental ΔAUC toward zero.

2. The labels are simulator output rather than clinical outcomes: the pre-registered openCARP ground truth was substituted with monodomain Mitchell–Schaeffer, a deviation logged as such.

3. The clinical endpoint (Section 4.2) is underpowered by construction, and its headline number is an artefact we report rather than a result we defend. On the observed 48/82 event rate, power at the $\Delta\text{AUC} \geq 0.05$ gate used for the simulator endpoint is 0.24 at best and 0.13 under a weaker shared-noise assumption, so that gate is *not* reused there. The minimum detectable effect is $\Delta\text{AUC} \approx 0.10$ to 0.15, and 80% power at 0.05 would need 371 to 859 patients (`results/power_real_outcomes.json`). We therefore do *not* report the clinical null as evidence that SFI adds nothing to recurrence prediction; the study cannot resolve an effect of the size that would matter. A further limitation is specific to this cohort and was disclosed before the analysis: UW fibrosis is categorical rather than continuous, which weakens the very baseline SFI must beat, and the baseline duly came out below chance. A cleaner test of the clinical question would require a cohort with continuous fibrosis imaging and several hundred patients, which this one is not.

4. Our first pass over the UW release was wrong in two ways, detailed with their controls in Section 4.5: a fibre field that is a constant vector in all 164 meshes, which is a placeholder rather than data (fibre orientations are IRB-withheld and meant to be regenerated), and element tag 164, the caps over the atrial orifices, which we had been treating as half-fibrotic tissue on an undocumented release. Both readings have since been corrected against the cohort authors' own tag legend. Repairing both raises inducibility only from 6/82 to 14/82, and none of the three substrate explanations for the cohort's low rate is supported by direct control; Control 4 attributes the low rate instead to a fibrosis burden about 25% lower on the meshes it simulates (0.248 against Roney's 0.333), reproduced by swapping the burden in either direction. Its regions also rest on a PCA surrogate rather than registered atrial coordinates. The cohort's predictive test is uninformative about SFI, and what remains open is narrower: openCARP returns 4/82 here and the same rescaling does not recover it (0/30, Section 4.6).

5. The inducibility label is stable in aggregate but noisy per subject, which attenuates every ΔAUC we report. Destroying the Roney fibre field holds the rate at 20/62 yet flips 10/62 individual verdicts, and repairing the UW substrate retains only 2 of its 6 original positives. That compounds the mesh-convergence instability above, where just 7/24 subjects were consistent across six resolutions. Non-differential label noise biases incremental AUC toward zero, so the null we report is partly a statement about the labeller and not only about SFI.

6. The real cohorts involved $\sim$30 DeLong comparisons, and the pre-registration fixes a multiplicity plan only for the clinical family of §8 (fixed-sequence gatekeeping, so that its primary is tested at the full $\alpha = 0.05$) and none for this simulator family, so every p-value we report here is unadjusted; on the full 182-subject corrected analysis none reaches the pre-registered $\alpha = 0.05$. The smallest is $p = 0.121$ (`fibrosis_vs_+SFI`, gradient boosting, UW), and that comparison sits on a base AUC of 0.562 with eight positives, so it is uninformative rather than borderline. An earlier version reported this family's combined estimate as the study's one positive at $p = 0.009$; it is withdrawn (Section 5.1).

7. ρ^* and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

8. **No sociodemographic characterisation of the clinical cohort is possible from the released data.** Both cohorts are de-identified public deposits of anatomy: they ship anatomical fields only — a tissue or image-intensity field, plus fibre directions and universal atrial coordinates in the Roney case — and no age, sex, race, ethnicity, comorbidity or socioeconomic variable accompanies either release. Neither deposit contains an outcome label — the two-year recurrence outcomes were withheld from the public download and obtained by direct request to the cohort's authors (Section 4.2). We therefore cannot describe the clinical cohort demographically, cannot report model performance within sociodemographic subgroups, and cannot assess whether discrimination differs across them — the fairness and health-inequality items of the reporting guideline are unanswerable here rather than overlooked. This also means the representativeness of these patients relative to any target ablation population is unknown, and it is a limitation of the deposits rather than of the analysis: obtaining those variables would require returning to the originating centres under their own governance.

9. **There was no patient or public involvement.** This is a secondary computational analysis of previously published, de-identified, publicly deposited datasets, with no participant contact at any stage and no mechanism through which patients or the public could have shaped the question, the endpoint, or the reporting. We record the absence rather than leave it to inference, and note that the endpoint definition itself — documented recurrence after a blanking period — was set by the originating study, not by us and not with patient input at this stage.

7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features, either at synthetic scale or on 182 patients of real anatomy, where the combined estimate is $+0.012$ with an interval that excludes the pre-registered gate. The mechanism decomposes, and only half of it is redundancy: the aggregate SFI summaries are reconstructible from the competitor set at $R^2 = 0.970$, while the extreme and edge-level columns carry variance of their own that does not predict the label. Independently of the predictive question, the single-vector estimator is ill-conditioned on real tissue ($\rho \gg 1$), and lawfully more so as meshes are refined.

The same conclusion holds on real clinical outcomes, with a warning attached. The pre-registered endpoint on two-year recurrence in the 82 UW patients returned $\Delta\text{AUC} = +0.197$ at $p = 0.0061$, past both of its thresholds, and is nevertheless not met: the baseline it improves on is anti-predictive at AUC 0.250, no feature on either side separates the outcome univariately, and nine columns of random noise reproduce the increment. The analysis plan had committed in writing to that reading before the outcomes were joined.

The core result is a carefully scoped boundary rather than a discovery: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily standard methodology would have reported a working biomarker. It would have done so twice from this project's own pipeline: once in silico, at $\Delta\text{AUC} = +0.103$ and $p = 0.009$, withdrawn on more data; once on real patients, at $\Delta\text{AUC} = +0.197$ and $p = 0.0061$, caught by a guard written before the data arrived. What the study cannot settle is the absolute inducibility rate, which does not converge within the mesh resolutions we could afford; the comparative claims, which score identical labels in both arms, do not depend on it. Whether the null is an artefact of the stand-in labeller is addressed directly rather than inferred from concordance: re-scoring the same comparison against an independent solver's verdicts leaves it in place, with a combined

increment of +0.028 and no arm clearing the pre-registered threshold. That re-scoring is an exploratory check, since the pre-registration bars those labels from any endpoint until their calibration gate passes on both cohorts, which it does not (Section 4.6).

A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel, an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals $\mathbb{Q}$, so `Print Assumptions` reports "Closed under the global context" for every theorem: zero axioms. A mathematically identical companion over the reals $\mathbb{R}$ (`formal/sfi_edge_identity.v`) inherits Coq's two standard real-number-library axioms and is kept only to show that the statements hold over $\mathbb{R}$.

Table 5: Kernel-checked theorems ($v = e_i - e_j$ the edge vector, $E_{ij} = \partial L/\partial w_{ij} = vv^\top$). T1–T5 are axiom-free over $\mathbb{Q}$.

name	statement	meaning
`T1_rank_one_quadratic_form`	$\varphi^\top(vv^\top)\varphi = (v\cdot\varphi)^2$	quadratic form of a rank-one edge matrix
`dot_edge` (T2)	$(e_i - e_j)\cdot\varphi = \varphi_i - \varphi_j$	the edge vector selects the potential difference
`T3_sfi_edge_sensitivity`	$\varphi^\top E_{ij}\varphi = (\varphi_i - \varphi_j)^2$	**the SFI edge term**
`T4_sfi_term_nonneg`	$0 \le w \Rightarrow 0 \le w(\varphi_i - \varphi_j)^2$	each SFI contribution is non-negative
`T5_aggregate_nonneg`	$\sum(\text{nonneg terms}) \ge 0$	SFI is a Dirichlet energy; the Laplacian form is PSD

Combined with the classical Hellmann–Feynman theorem $\partial\lambda_2/\partial w_{ij} = \varphi_2^\top(\partial L/\partial w_{ij})\varphi_2$ (Ghosh–Boyd 2006 [2], an analysis result we cite rather than re-prove), T3 is exactly the closed form $\partial\lambda_2/\partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$. Lean/Mathlib was attempted, but its toolchain binaries are distributed only as GitHub release assets, which the compute environment's egress policy blocks with a hard 403; Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

B Reproducibility

All stochastic steps are seeded, and the frozen configuration behind every reported number is fixed as module-level constants in the source rather than supplied at run time. Commit provenance is carried by the repository history rather than stamped into the result files. Results in this paper were produced on Python 3.12.10 with NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0, pandas 3.0.5 and Matplotlib 3.11.1; the openCARP cross-check used openCARP v19.0. The per-subject record cache is keyed by a hash of the frozen configuration, so changing any frozen constant invalidates the cache rather than silently reusing stale labels. The numeric verification is reproduced with `python scripts/verify_sfi_identity.py` and `python scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks), the convergence analysis with `python scripts/convergence_analyze.py roney`, the clinical endpoint with `python scripts/clinical_endpoint.py`, and the figures with `python scripts/make_paper_figures.py`. The datasets are the Roney LA cohort (Zenodo 5801337) and the UW/Boyle meshes (Dryad `10.5061/dryad.kkwh70sg0`), both of which are public and de-identified. The two-year recurrence outcomes used in Section 4.2 were supplied by the UW cohort's authors, who confirm they carry no confidentiality restriction; they are available from that group, and 15 of the 82 appear

in the source publication. The LGE intensity ratios and fibre orientations for that cohort are withheld under its IRB and are not obtainable; fibres can be regenerated with the tools that study cites.

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